

Customer Churn Prediction Using Machine Learning

Course: AI for Engineers (UCS321)

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Abstract

Customer churn prediction is a critical business problem in the telecom industry. This project presents a machine learning-based approach to proactively predict whether a customer is likely to leave the service provider. The primary objective was to develop a predictive model with high recall to minimize false negatives. Multiple algorithms were evaluated, and the Random Forest classifier was selected as the optimal solution due to its robustness and superior performance.

1. Introduction

Telecom companies face substantial revenue losses due to customer churn. Traditional retention systems are reactive and inefficient. This project aims to build an intelligent predictive system that identifies at-risk customers before cancellation.

2. Problem Statement

The objective is to develop an automated machine learning system capable of predicting customer churn based on demographic and service usage data. The system must minimize false negatives and support proactive retention strategies.

3. Project Objectives

- Develop an accurate churn prediction model.
- Optimize recall to reduce false negatives.
- Compare Logistic Regression, Decision Tree, and Random Forest algorithms.

4. Dataset Overview

Source: Telecom Customer Churn Dataset

Total Samples: 7,043

Total Features: 21

Target Variable: Churn (Binary Classification)

Data Types: Combination of categorical and numerical features
Challenge: Class imbalance with more non-churners than churners.

5. Data Preprocessing

- Missing values handled using median imputation.
- Categorical variables encoded using Label Encoding.
- Numerical features standardized.
- Dataset split into 80% training and 20% testing.

6. Exploratory Data Analysis (EDA)

EDA revealed strong relationships between tenure, contract type, monthly charges, and churn probability. A correlation matrix was used to identify multicollinearity.

7. Model Selection and Justification

Three models were evaluated:

1. Logistic Regression – Baseline model.
2. Decision Tree – Captures non-linear relationships but prone to overfitting.
3. Random Forest – Ensemble method reducing variance and improving generalization.

8. Model Training Process

- Baseline model trained using Logistic Regression.
- Random Forest optimized using GridSearchCV with 5-fold cross-validation.
- Evaluation performed on unseen test data using Recall and F1-score.

9. Evaluation Metrics

Accuracy – Overall correct predictions.

Precision – Minimizes unnecessary marketing costs.

Recall – Critical for identifying actual churners.

F1-Score – Balance between precision and recall.

10. Results and Performance

The Random Forest model achieved approximately 80–82% accuracy with the highest recall among tested models. Feature importance analysis identified tenure and contract type as major predictors.

11. Challenges and Solutions

- Class imbalance addressed using class-weight adjustments.
- Overfitting mitigated using Random Forest and depth control.
- Hyperparameter tuning performed using GridSearchCV.

12. Business Applications

- Proactive retention campaigns.
- Targeted marketing optimization.
- Customer Lifetime Value risk assessment.

13. Future Scope

- Implementation of Deep Learning models (LSTM).
- Deployment as API using Flask or Streamlit.
- Real-time streaming integration using Apache Kafka.

14. Conclusion

This project successfully developed an end-to-end churn prediction pipeline. The Random Forest classifier provided the best balance of accuracy and recall. The system is scalable and ready for deployment in real-world telecom environments.